SYED WASAM JAFFRI

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SUMMARY

AI Engineer with hands-on experience building LLM-powered backends, RAG pipelines, and end-to-end ML systems for production. Proficient in FastAPI, FAISS, HuggingFace, and prompt engineering, with practical deployment via Docker and GitHub Actions CI. Track record of shipping production-grade AI applications from data ingestion through model serving and drift monitoring.

TECHNICAL SKILLS

PProgramming & Backend: Python, Java, C, REST API development (FastAPI, Uvicorn), asynchronous programming, OOP
LLM & NLP: RAG pipeline design, FAISS vector indexing, BAAI/bge embeddings, Flan-T5 (3B params), prompt engineering, semantic chunking, LLM performance evaluation (latency, accuracy, cost)
Machine Learning & AI: Supervised learning (scikit-learn), XGBoost, LightGBM, model calibration, threshold optimization, feature engineering & selection, SHAP explainability, ROC-AUC / PR-AUC / lift@k, computer vision (face encodings, PCA)

MLOps & Deployment: Docker, docker-compose, GitHub Actions CI, FastAPI model serving, Streamlit dashboards, model registry patterns, MLflow / Evidently (familiar), PSIbased drift monitoring
Data & Tools: pandas, NumPy, PyArrow, joblib, pdfplumber, python-docx, SQLite, MySQL, CSV/Parquet, data ingestion pipelines
Web: HTML, CSS, JavaScript, Streamlit, React (basic)
Quality & Observability: pytest, test-driven development, pre-commit (black, ruff, isort), mypy, structured logging
AI Platforms & Libraries: Hugging Face, FAISS, SHAP, LightGBM, Flan-T5, BAAI embeddings, Pydantic
Tools: Git / GitHub, virtualenv/pip, YAML config, Linux/Windows CLI

WORK EXPERIENCE

FlyrankAI — Backend AI Engineer Jul 2026 – Oct 2026 | Karachi, Pakistan

- Integrated vector databases for semantic search and knowledge retrieval, improving query relevance across production backend AI services
- Optimized prompt templates to reduce token usage and API costs through structured output constraints and few-shot compression
- Evaluated LLM performance across latency, accuracy, and cost dimensions; benchmarking findings directly informed model selection decisions
- Integrated external APIs, databases, and AI models into backend services built on FastAPI and Python
- Improved AI response quality and consistency through iterative prompt engineering and workflow optimization

EDUCATION

Bachelor of Artificial Intelligence | Oct 2023 – Present DHA Suffa University, Karachi, Pakistan
CGPA: 3.1

PROJECTS more on https://github.com/wasam0110

RAG Document Cataloger

https://github.com/wasam0110/Rag-Document-Cataloger Python, FastAPI, FAISS, HuggingFace (BAAI/bge, Flan-T5), SQLite, JWT, HTML/CSS/JavaScript

- Built production-ready RAG system supporting multi-format document ingestion (PDF, DOCX, TXT, CSV)
- Implemented semantic chunking with overlap and FAISS vector indexing for fast, context-aware document retrieval.
- Integrated BAAI/bge-base embeddings (768-dim) and google/flan-t5-xl (3B params) for high-quality AI-generated answers with automatic citations.
- Developed secure email-based user authentication using JWT tokens, bcrypt, and SMTP verification with session management.
- Designed a responsive frontend with an inline PDF viewer, dynamically detected document sections, and real-time loading indicators.

WearHowZ — Multi-Vendor Streetwear Marketplace (In Progress)

github.com/wasam0110/DripMarketplace Python, FastAPI, PostgreSQL/Supabase, JWT RS256, Docker, Next.js 15

- Built a layered async FastAPI backend (router → service → repository) with PostgreSQL/Supabase, Alembic migrations, JWT RS256 + Google OAuth, and role-based access for buyers, sellers, and admins
- Integrated PayFast, and full COD flow, payment timeout handling, guest order tokens, and IPN/webhook callbacks
- Engineered a 15% commission wallet system with 3-day hold periods, configurable withdrawal limits (PKR 500–200K), and background workers for wallet and order reconciliation
- Full integration and unit test coverage (orders, payments, returns, commission logic); Dockerized with Supabase Storage, Resend email, WhatsApp notifications, and Sentry monitoring
- Next.js 15 frontend in development (Zustand, TanStack Query)

Churn Prediction Platform

https://github.com/wasam0110/ChurnPredictionSystem Python, scikit-learn, XGBoost, LightGBM, FastAPI, Streamlit, SHAP, Docker, pytest

- Built an end-to-end churn prediction platform with automated data ingestion, 30+ engineered features, multi-model training (LR, RF, XGBoost, LightGBM), probability calibration, and cost-sensitive threshold optimization to maximize business profit over raw accuracy
- Integrated SHAP-based global and local explainability with what-if simulation and a next-best-action engine ranking 15 ROI-optimized retention actions per at-risk customer across 5 categories
- Deployed real-time FastAPI service (/predict, /predict/batch, /whatif) with Pydantic validation; 6-page Streamlit dashboard covering KPIs, model performance, and PSI-based drift monitoring

EXTRA CURICULAR

- **RUNNER-UP:** 15TH DSU Inter-Department Box Cricket Championship 2025 (CAPTAIN)
- Badminton
- Table tennis
- Football
- GYM